

Adaptive State-of-Health Forecasting for Second-Life LFP Cells Using Physics-Informed Synthetic Data and Hybrid Model Selection

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Second-life reuse of retired lithium iron phosphate (LFP) cells requires fast prediction of future degradation for each incoming cell. Instead of assuming a single degradation model is optimal for all cells, we develop an adaptive forecasting framework that selects between simple empirical extrapolation and a PyBaMM-trained neural operator based on early-cycle degradation complexity.

Approach (Figure 1). For each target cell the pipeline uses: (1) a short diagnostic characterisation together with the first $K=50$ operating cycles as early-cycle context; (2) a PyBaMM [1] SPM corpus on Prada2013 LFP chemistry [2] with perturbed degradation trajectories around seven anchor cells from three suppliers (CALB, EVE, REPT), providing training trajectories that cover the second-life SoH range; (3) a Neural ODE [3] operator trained using PyBaMM-generated degradation trajectories and degradation-model-derived features, forecasting the residual SoH trajectory from the diagnostic features and the $K=50$ context window; (4) an exponential $SoH(n)=a+be^{-cn}$ baseline fitted on the same context window; and (5) a residual-based selector that predicts the last 20% of the context from a fit on the first 80% and picks the neural operator whenever the context-holdout residual exceeds a predeclared threshold τ , chosen on a synthetic-only selector pool disjoint from the external cells.

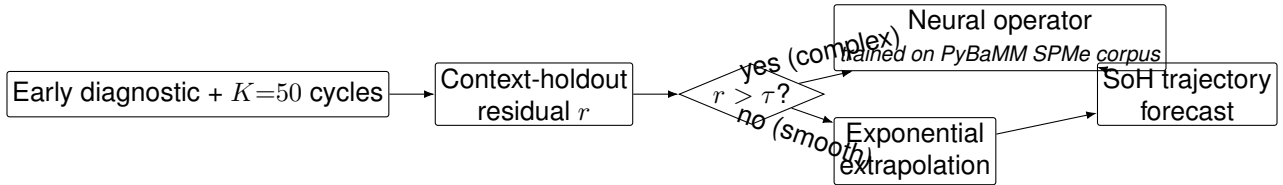


Figure 1: Adaptive selector between empirical exponential forecasting and a neural operator trained on physics-informed synthetic degradation trajectories. A context-holdout residual measures early-cycle degradation complexity; the predeclared threshold τ is chosen on synthetic trajectories, not on the external cells.

Results. Across three external LFP cells excluded from operator training (CALB_0029, EVE_0003, REPT_0028), exponential forecasting achieved 0.70 pp mean RMSE, while the physics-informed operator achieved 0.80 pp. A context-residual selector combined the complementary models and reduced mean RMSE to 0.47 pp, matching the oracle model choice on the evaluated cells. The residual-based selector reproduced the oracle model choice on the three evaluated external cells: exponential for the two smoother trajectories (CALB_0029, REPT_0028) and the neural operator for the noisier EVE_0003. The exponential baseline and the operator show complementary strengths – exponential wins on smooth fade while the operator wins on the more nonlinear trajectory – motivating selection rather than committing to either model alone.

Table 1: External held-out SoH RMSE (percentage points) on three cells excluded from operator training. The hybrid row uses the full- θ operator when the residual-based selector chooses it.

Method	CALB_0029	EVE_0003	REPT_0028	Mean
Exponential	0.44	1.54	0.13	0.70
Full- θ operator	1.06	0.84	0.51	0.80
No- θ operator	1.08	0.89	0.49	0.82
Reduced- θ operator	1.18	0.97	0.55	0.90
Hybrid ($\tau=0.54$ pp)	0.44	0.84	0.13	0.47

Ablations indicate that PyBaMM-generated trajectories provide useful training diversity, while explicit degradation-parameter conditioning provides limited additional benefit.

The current evaluation is limited to three external cells, and longer-horizon projections require additional experimental validation.

- [1] V. Sulzer et al., *J. Open Res. Software* **9** (2021), 14. [2] M. Prada et al., *J. Power Sources* **219** (2012), 353–362. [3] R. T. Q. Chen et al., *NeurIPS* (2018).